02_eda

March 28, 2026

[1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

[2]:
```
df = pd.read_csv("../data/zomato_cleaned.csv")
```

[3]:
```
print("Dataset Shape:", df.shape)
print(df.head(3))
```

```
Dataset Shape: (3448, 22)
                   name                                                 url  \
0        Sri Udupi Park  https://www.zomato.com/bangalore/sri-udupi-par…
1         Meghana Foods  https://www.zomato.com/bangalore/meghana-foods…
2  Donne Biriyani House  https://www.zomato.com/bangalore/donne-biriyan…

                                          cuisines                    area  \
0  south indian, north indian, chinese, street fo…  Indiranagar, Bangalore
1            biryani, andhra, north indian, seafood  Indiranagar, Bangalore
2                                          biryani  Indiranagar, Bangalore

                timing                                       full_address  \
0   7am - 11pm (Today)  273, Monalisa, 6th Main, 100 Feet Road, Indira…
1      Opens at 6:30pm  544, First Floor, CMH Road, Near Indiranagar M…
2  11am - 11pm (Today)  8/ 9, 17th F Cross, 2nd Stage, Indiranagar, Ba…

     phonenumber  ishomedelivery  istakeaway  isindoorseating  …  \
0  +919945977774               1           1                1  …
1  +918041135050               1           1                1  …
2  +918861564169               1           1                1  …

   dinner_reviews  delivery_ratings  delivery_reviews               knownfor  \
0             462               4.1             16000                    NaN
1            1654               4.3             28600  Spicy Chicken Biryani
2             411               3.5             33200                    NaN

                                       popular dishes  \
0  Filtered Coffee, Sambhar, Pav Bhaji, Gobi Manc…
```

```
1  Authentic Hyderabadi Biryani, Paneer Biryani, …
2                                                NaN

                                     peopleknownfor averagecost       city  \
0  Economical, Prompt Service, Hygiene, Quality F…         450  Bangalore
1  Boneless Chicken Biryani, Ample Seating Area, …         700  Bangalore
2  Great Recommendations, Nice Taste, Great Ambia…         300  Bangalore

   latitude longitude
0       NaN       NaN
1       NaN       NaN
2       NaN       NaN

[3 rows x 22 columns]
```

[4]:
```python
print(df.describe())
```

```
       ishomedelivery   istakeaway  isindoorseating    isvegonly  \
count     3448.000000  3448.000000      3448.000000  3448.000000
mean         0.999130     0.977088         0.921404     0.118910
std          0.029488     0.149644         0.269147     0.323729
min          0.000000     0.000000         0.000000     0.000000
25%          1.000000     1.000000         1.000000     0.000000
50%          1.000000     1.000000         1.000000     0.000000
75%          1.000000     1.000000         1.000000     0.000000
max          1.000000     1.000000         1.000000     1.000000

       dinner_ratings  dinner_reviews  delivery_ratings  delivery_reviews  \
count     3448.000000     3448.000000       3448.000000       3448.000000
mean         3.636688      366.133991          3.932338       3626.352088
std          0.471438     1056.205709          0.277627       6997.316540
min          2.000000        0.000000          2.600000          0.000000
25%          3.300000       20.000000          3.800000        310.750000
50%          3.700000       85.000000          4.000000       1090.500000
75%          4.000000      268.000000          4.100000       3747.250000
max          4.900000    26500.000000          4.600000      81500.000000

       averagecost  latitude  longitude
count  3448.000000       0.0        0.0
mean    504.886891       NaN        NaN
std     308.726271       NaN        NaN
min     100.000000       NaN        NaN
25%     300.000000       NaN        NaN
50%     400.000000       NaN        NaN
75%     600.000000       NaN        NaN
max    1900.000000       NaN        NaN
```

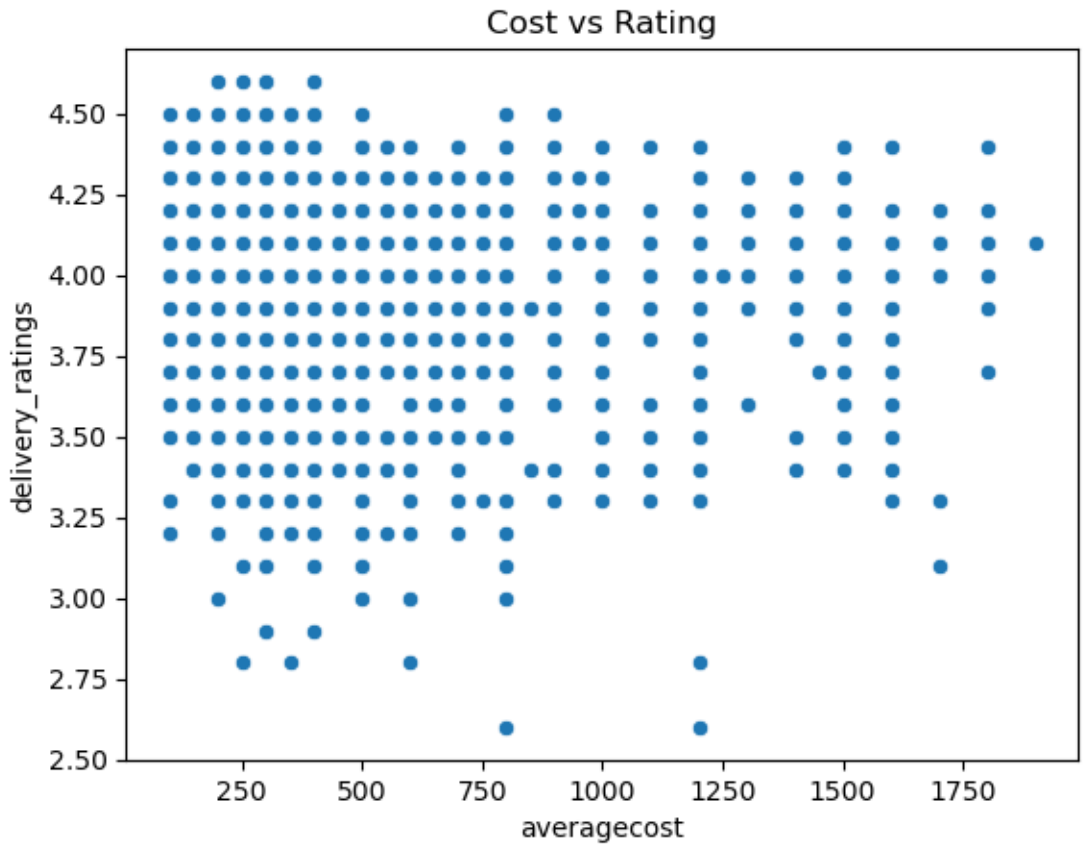
[5]:
```
area_competition = df.groupby("area")["name"].count().
 ↪sort_values(ascending=False)
print(area_competition.head(10))
```

```
area
HSR, Bangalore                187
Indiranagar, Bangalore        151
Electronic City, Bangalore    144
BTM, Bangalore                137
JP Nagar, Bangalore           136
Whitefield, Bangalore         130
Marathahalli, Bangalore       124
Jayanagar, Bangalore          120
New BEL Road, Bangalore        94
Kalyan Nagar, Bangalore        88
Name: name, dtype: int64
```

[6]:
```
df["total_reviews"] = df["dinner_reviews"] + df["delivery_reviews"]

top_demand = df.sort_values("total_reviews", ascending=False).head(10)

print(top_demand[["name","area","total_reviews"]])
```

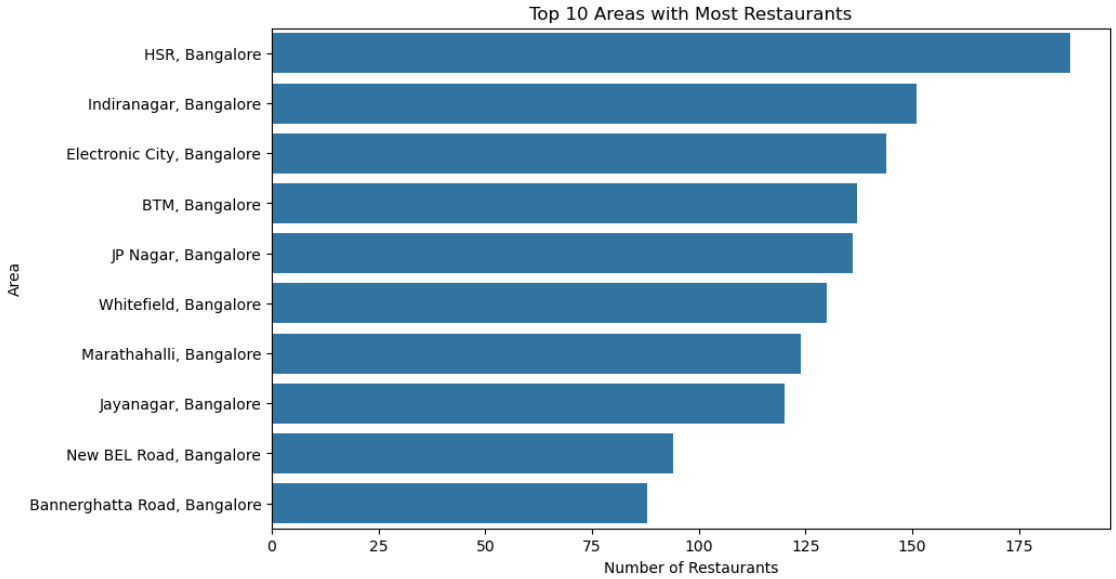
```
                        name                             area  total_reviews
337        Imperio Restaurant            Brookefield, Bangalore          84124
3126        Empire Restaurant            Indiranagar, Bangalore          81800
862   Falahaar & Kota Kachori  Koramangala 7th Block, Bangalore          80930
829         Empire Restaurant  Koramangala 5th Block, Bangalore          79917
846             Meghana Foods  Koramangala 5th Block, Bangalore          76728
2452       Imperio Restaurant           Kalyan Nagar, Bangalore          74113
1414          Ayodhya Upachar           Banashankari, Bangalore          72238
40                    Al Daaz                    HSR, Bangalore          65023
1380            Meghana Foods               Jayanagar, Bangalore          62895
878              Asha Tiffins                    HSR, Bangalore          61188
```

[7]:
```
sns.scatterplot(x="averagecost", y="delivery_ratings", data=df)
plt.title("Cost vs Rating")
plt.show()
```

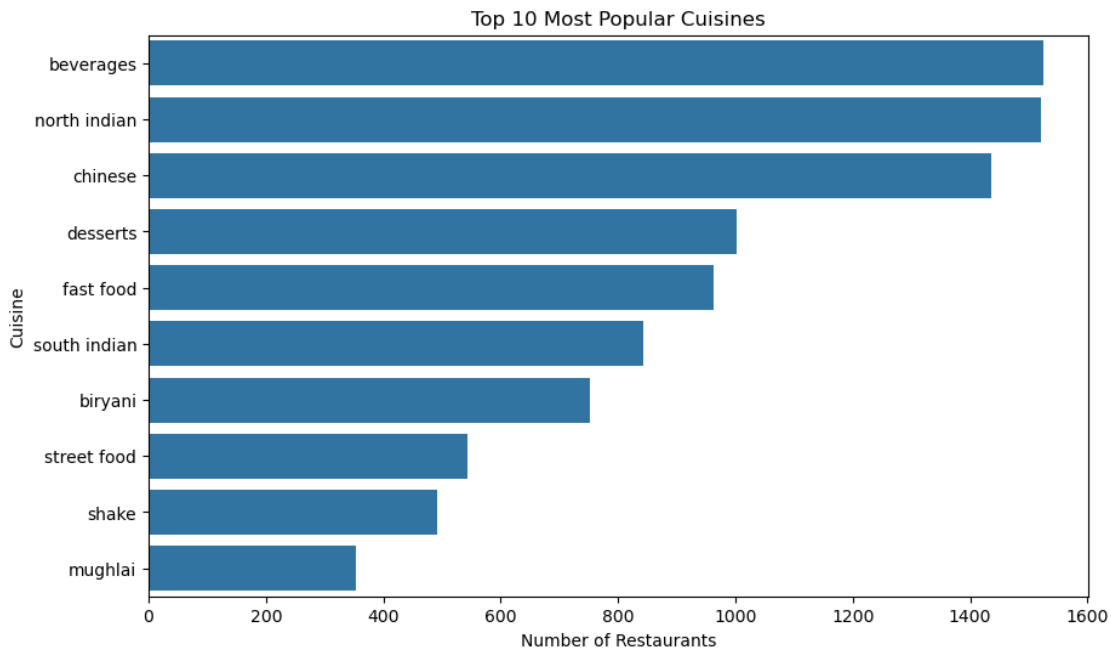


[8]:
```
area_counts = df["area"].value_counts().head(10)
```

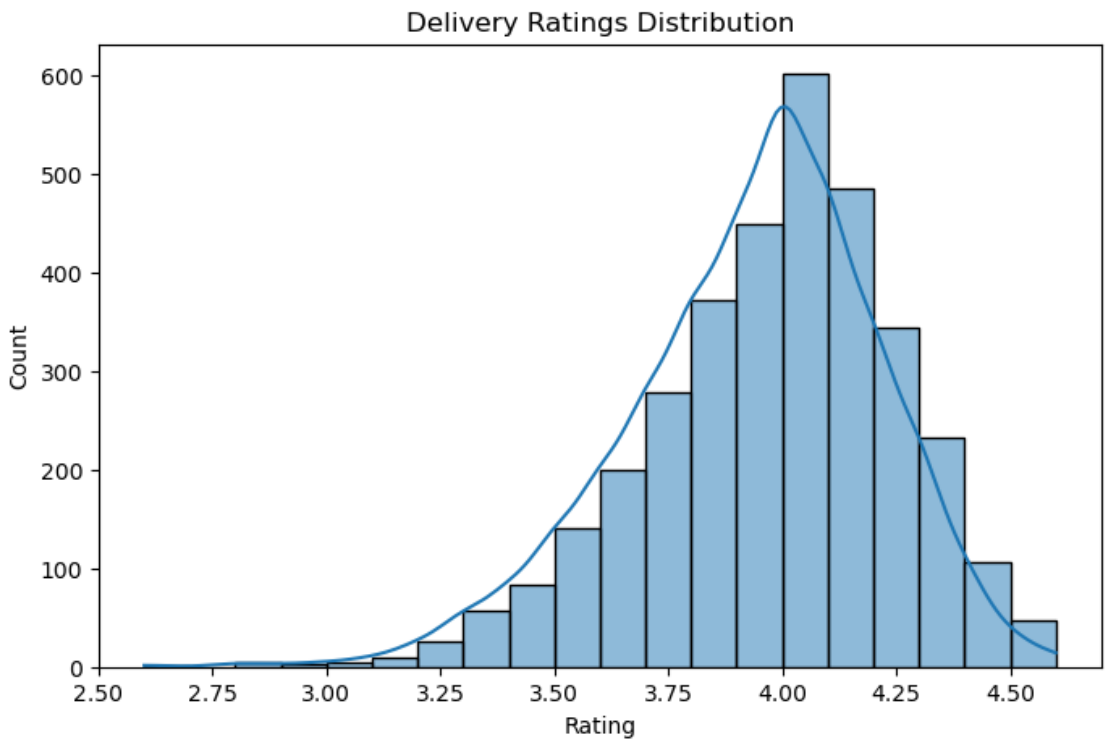
[9]:
```
plt.figure(figsize=(10,6))
sns.barplot(x=area_counts.values, y=area_counts.index)
plt.title("Top 10 Areas with Most Restaurants")
plt.xlabel("Number of Restaurants")
plt.ylabel("Area")
plt.show()
```



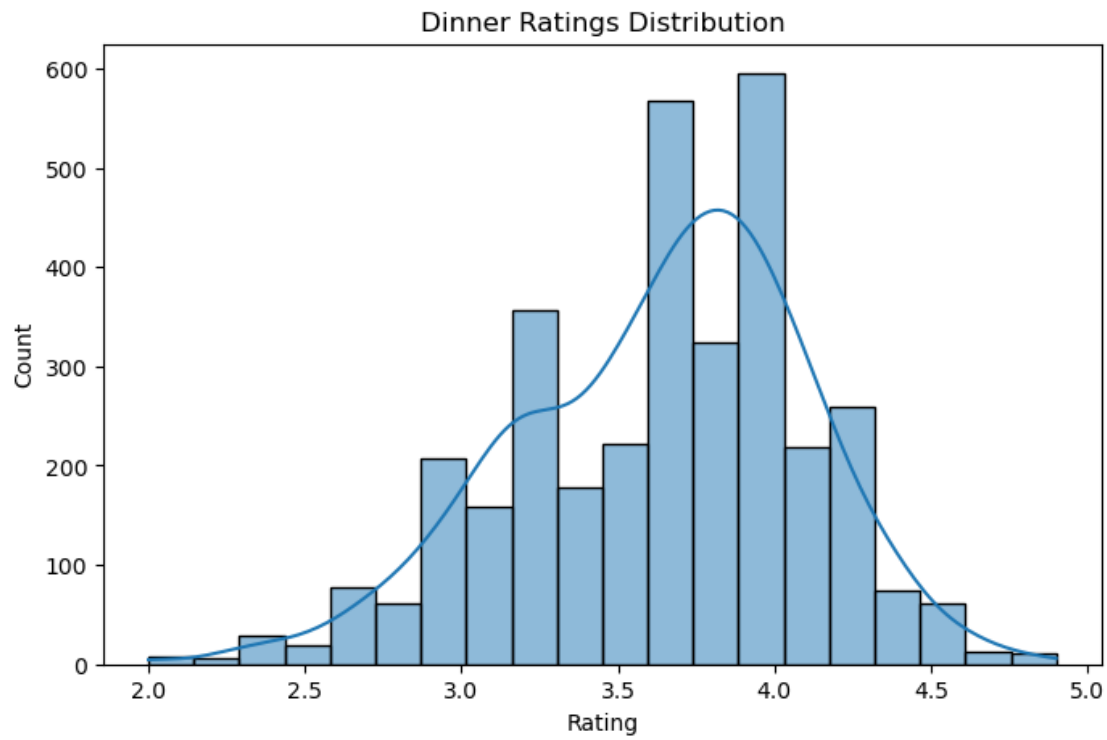
[10]:
```
df["cuisines"] = df["cuisines"].astype(str)
df["cuisines"] = df["cuisines"].str.split(",")

cuisine_df = df.explode("cuisines")
cuisine_df["cuisines"] = cuisine_df["cuisines"].str.strip()

top_cuisines = cuisine_df["cuisines"].value_counts().head(10)

plt.figure(figsize=(10,6))
sns.barplot(x=top_cuisines.values, y=top_cuisines.index)
plt.title("Top 10 Most Popular Cuisines")
plt.xlabel("Number of Restaurants")
plt.ylabel("Cuisine")
plt.show()
```



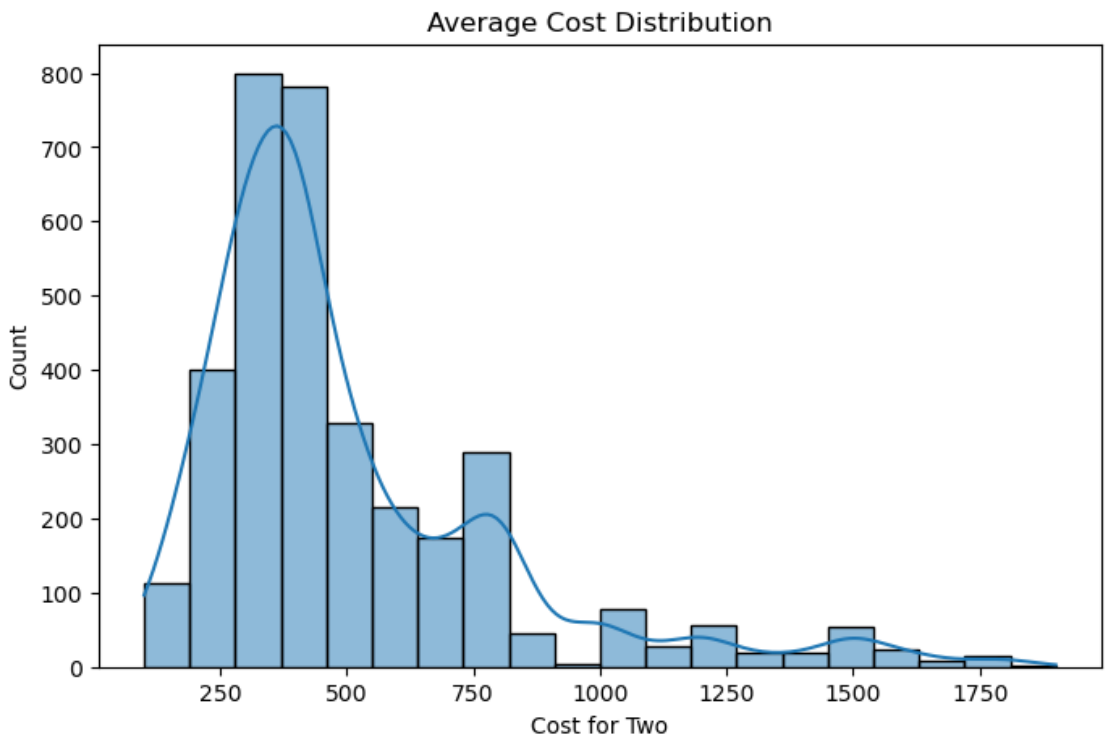
[11]:
```
plt.figure(figsize=(8,5))
sns.histplot(df["delivery_ratings"], bins=20, kde=True)
plt.title("Delivery Ratings Distribution")
plt.xlabel("Rating")
plt.ylabel("Count")
plt.show()
```



[12]:
```
plt.figure(figsize=(8,5))
sns.histplot(df["dinner_ratings"], bins=20, kde=True)
plt.title("Dinner Ratings Distribution")
plt.xlabel("Rating")
plt.ylabel("Count")
plt.show()
```



[13]:
```
plt.figure(figsize=(8,5))
sns.histplot(df["averagecost"], bins=20, kde=True)
plt.title("Average Cost Distribution")
plt.xlabel("Cost for Two")
plt.ylabel("Count")
plt.show()
```

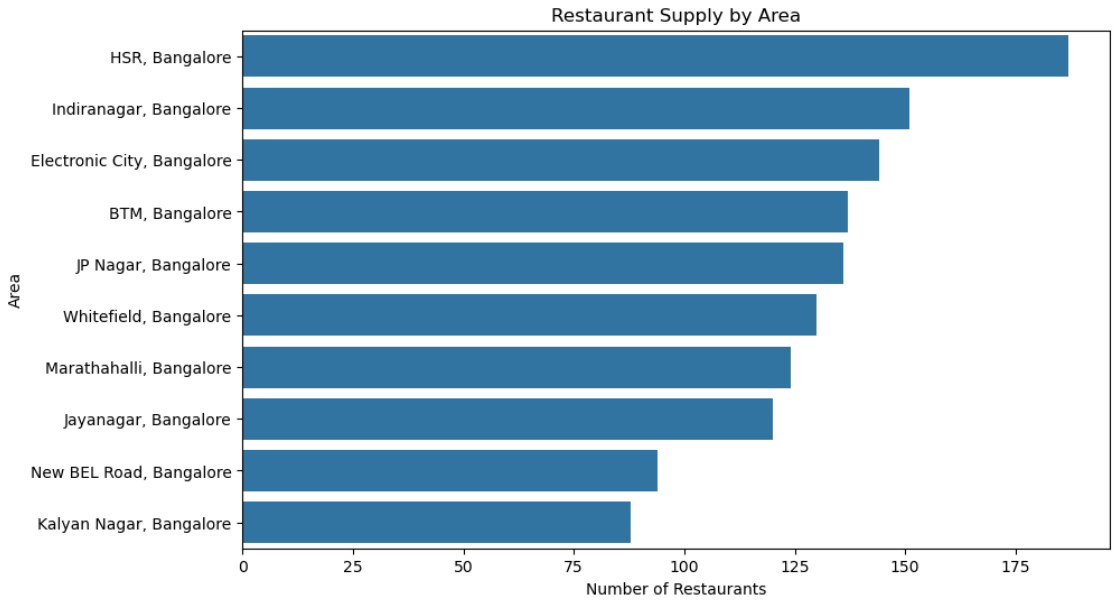


[14]:
```
df["total_reviews"] = df["dinner_reviews"] + df["delivery_reviews"]

top_demand = df.sort_values(by="total_reviews", ascending=False).head(10)

print(top_demand[["name","area","total_reviews"]])
```

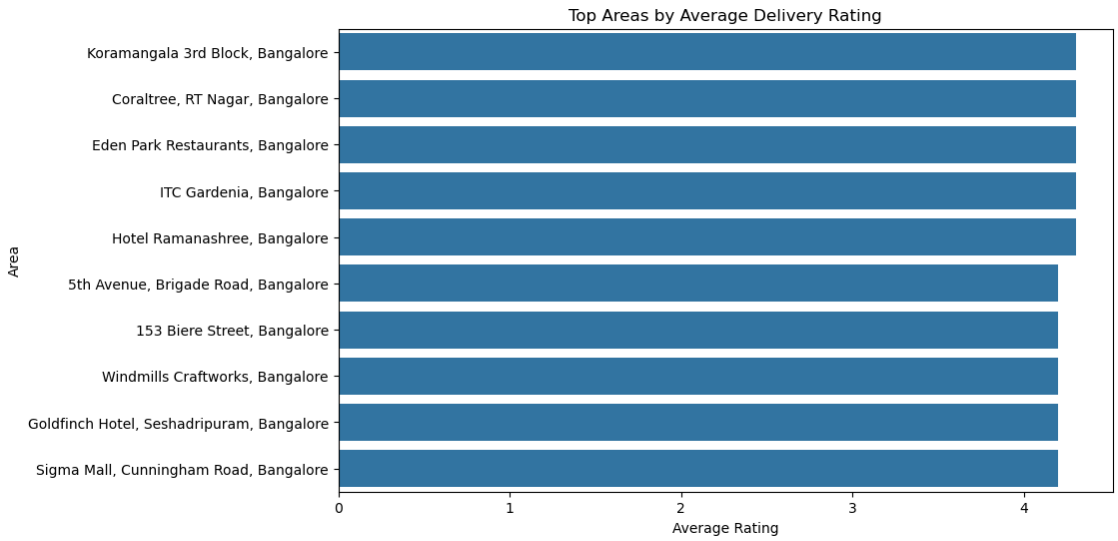
```
                        name                              area  total_reviews
337        Imperio Restaurant            Brookefield, Bangalore          84124
3126        Empire Restaurant            Indiranagar, Bangalore          81800
862   Falahaar & Kota Kachori  Koramangala 7th Block, Bangalore          80930
829         Empire Restaurant  Koramangala 5th Block, Bangalore          79917
846             Meghana Foods  Koramangala 5th Block, Bangalore          76728
2452       Imperio Restaurant           Kalyan Nagar, Bangalore          74113
1414          Ayodhya Upachar           Banashankari, Bangalore          72238
40                    Al Daaz                    HSR, Bangalore          65023
1380            Meghana Foods               Jayanagar, Bangalore          62895
878              Asha Tiffins                    HSR, Bangalore          61188
```

[15]:
```
restaurant_supply = df.groupby("area")["name"].count().
 ↪sort_values(ascending=False).head(10)

plt.figure(figsize=(10,6))
sns.barplot(x=restaurant_supply.values, y=restaurant_supply.index)
```

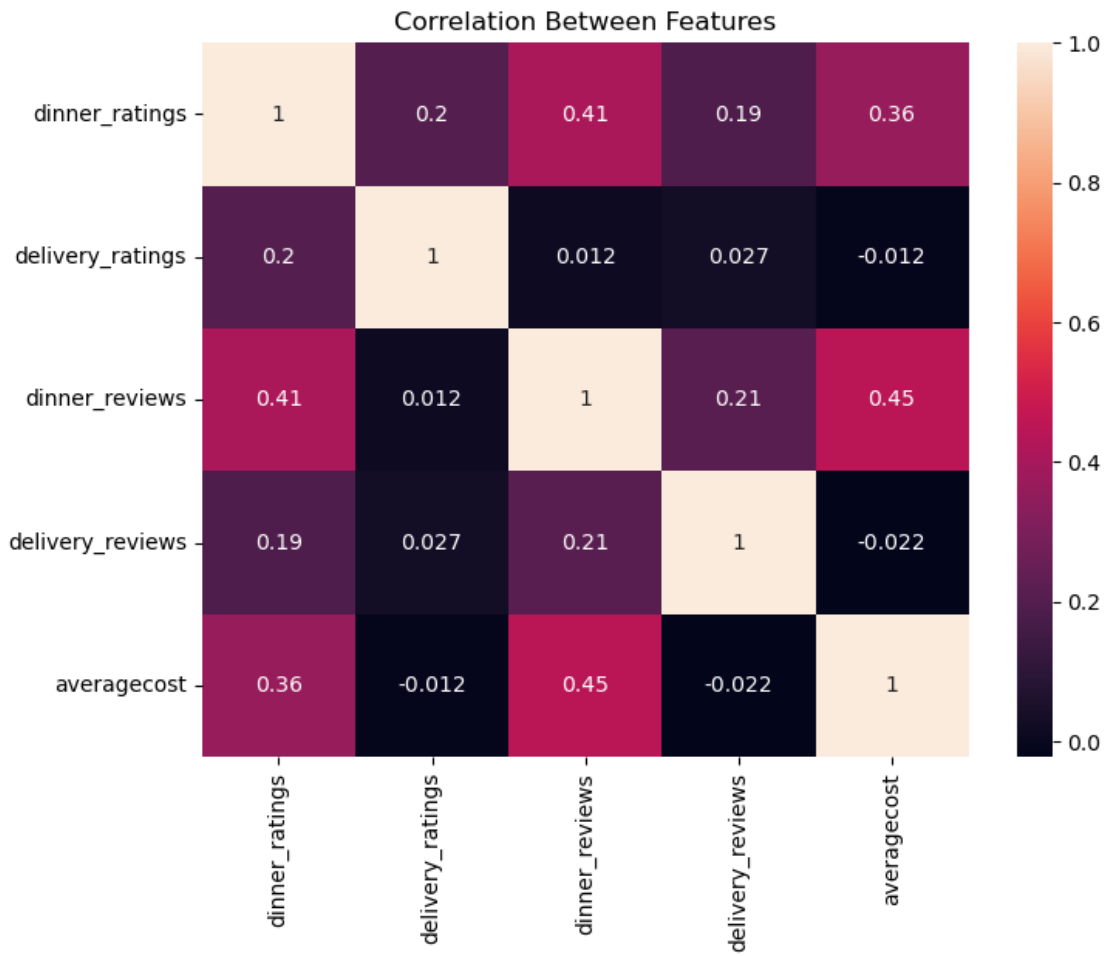
```
plt.title("Restaurant Supply by Area")
plt.xlabel("Number of Restaurants")
plt.ylabel("Area")
plt.show()
```



[16]:
```
area_rating = df.groupby("area")["delivery_ratings"].mean().
 ↪sort_values(ascending=False).head(10)

plt.figure(figsize=(10,6))
sns.barplot(x=area_rating.values, y=area_rating.index)
plt.title("Top Areas by Average Delivery Rating")
plt.xlabel("Average Rating")
plt.ylabel("Area")
plt.show()
```



[17]:
```
plt.figure(figsize=(8,6))
sns.heatmap(df[[
    "dinner_ratings",
    "delivery_ratings",
    "dinner_reviews",
    "delivery_reviews",
    "averagecost"
]].corr(), annot=True)

plt.title("Correlation Between Features")
plt.show()

print("EDA Completed")
```




```
EDA Completed
```